MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG5DSEBOT302, MG5DSEBBT300**
Course Title**: PLANT BIOTECHNOLOGY**

Time: **2 Hrs** Maximum Marks: **70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)
Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A: Very Short Answer Questions
Answer 10 questions
Each question carries 2 marks

	Questions	Taxonomy	CO
1.	Name two common sterilization methods used in plant tissue culture.	Understand	1
2.	Mention a feature of golden rice.	Remember	5
3.	What are edible vaccines?	Understand	5
4.	Differentiate cDNA library from genomic library.	Understand	3
5.	Name an organism involved in the bioremediation.	Understand	5
6.	What is synthetic biology?	Remember	5
7.	Define plasmid.	Understand	2
8.	Examine the peculiarities of climate resilient crops.	Remember	5
9.	Explain totipotency.	Understand	1
10.	Expand YAC.	Understand	2
11.	What is electroporation?	Remember	4
12.	What are molecular scissors?	Understand	2

(2 X 10 = 20)

Part B: Short answer Questions
Answer 8 questions
Each question carries 5 marks

13.	Write short notes on sterilization methods	Apply	1
14.	How transformants are screened by alpha-complementation?	Understand	3
15.	Describe any two applications of biotechnology.	Understand	5
16.	Differentiate electroporation from microinjection.	Analyse	4
17.	What are cybrids? How they are obtained?	Understand	1
18.	Describe the features of CRISPR- Cas9 gene editing tool.	Apply	5
19.	Explain the features of Ti plasmid.	Understand	2
20.	Examine the role of reporter genes in the selection of transgenic plants.	Apply	4
21.	How synthetic biology helps in plant metabolic engineering?	Understand	5
22.	Explain bacterial transformation.	Apply	3

(5 X 8 = 40)

Part C: Essay Questions
Answer 1 question
Each question carries 10 marks

23.	What is rDNA technology? Elaborate its steps.	Understand	3
24.	Explain the principle and procedure of micropropagation. Discuss how micropropagation contributes to the rapid multiplication of elite plant varieties.	Understand	1

(10 X 1 = 10)

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

Course Code: **MG5DSEBOT303, MG5DSEBBT301**

Course Title: **GREEN TECHNOLOGY AND SUSTAINABLE DEVELOPMENT**

Time: 2 Hrs. **Maximum Marks: 70**

Remember (K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

		Taxonomy	CO
1.	What is meant by green chemistry??	[U]	[1]
2.	Define sustainable development.	[K]	[3]
3.	Define carbon credits.	[K]	[2]
4.	What is meant by the Zero Waste concept??	[U]	[2]
5.	What is ISO 14001:2019?	[K]	[4]
6.	Define Environmental Impact Assessment.	[U]	[4]
7.	What is meant by renewable energy gadgets?	[K]	[1,5]
8.	What is Eco-mark certification??	[U]	[5]
9.	Define green technology.	[U]	[1]
10.	What is phyto-nanotechnology?	[C]	[2]
11.	What is meant by green belt?	[K]	[5]
12.	Mention any two benefits of green buildings.	[K]	[5]

[2×10=20]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **5** marks

13.	Explain the importance and applications of green technology.	[K]	[1]
14.	Describe the major green technology initiatives in India.	[U]	[1]
15.	Explain microwave-assisted synthesis and supercritical fluid extraction as green methods.	[U]	[1]
16.	Discuss Reduce, Reuse and Recycle as strategies for sustainable waste management.	[U]	[2]
17.	Describe bioremediation and analyse its significance in environmental management.	[An]	[2]
18.	Describe the scope and goals of Life Cycle Analysis.	[U]	[4]
19.	Discuss water conservation and water audit as green practices.	[K]	[5]
20.	Write a note on solar energy and wind energy as renewable energy sources.	[K]	[3]
21.	Discuss the social, economic and environmental dimensions of sustainability.	[K]	[3]
22.	Explain the role of governmental bodies in green planning.	[K]	[5]

[5×8=40]

Part C

Essay Questions

Answer any **1** question

Each question carries **10** marks

23.	Explain the twelve principles of green chemistry. Discuss the importance and applications of green technology in achieving sustainable development.	[K]	[1]
24.	Discuss LEED certification, Eco-mark certification in India, green cities, green belts and the role of governmental bodies in green planning.	[U]	[5]

[10×1=10]